

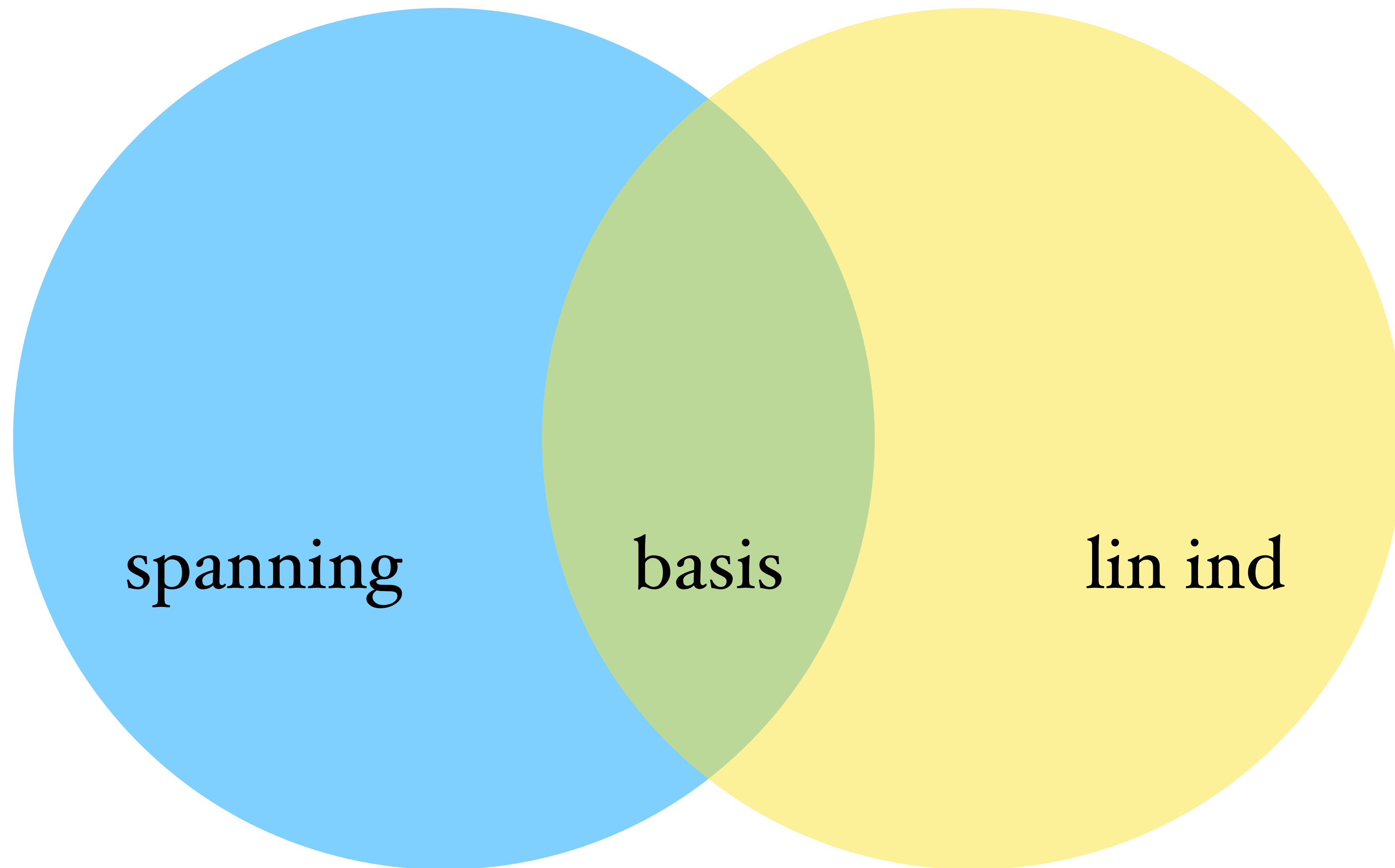
Abstract Linear Algebra

Prof. Lanier

jlancier8@uchicago.edu

Please pick up a copy of the handouts.

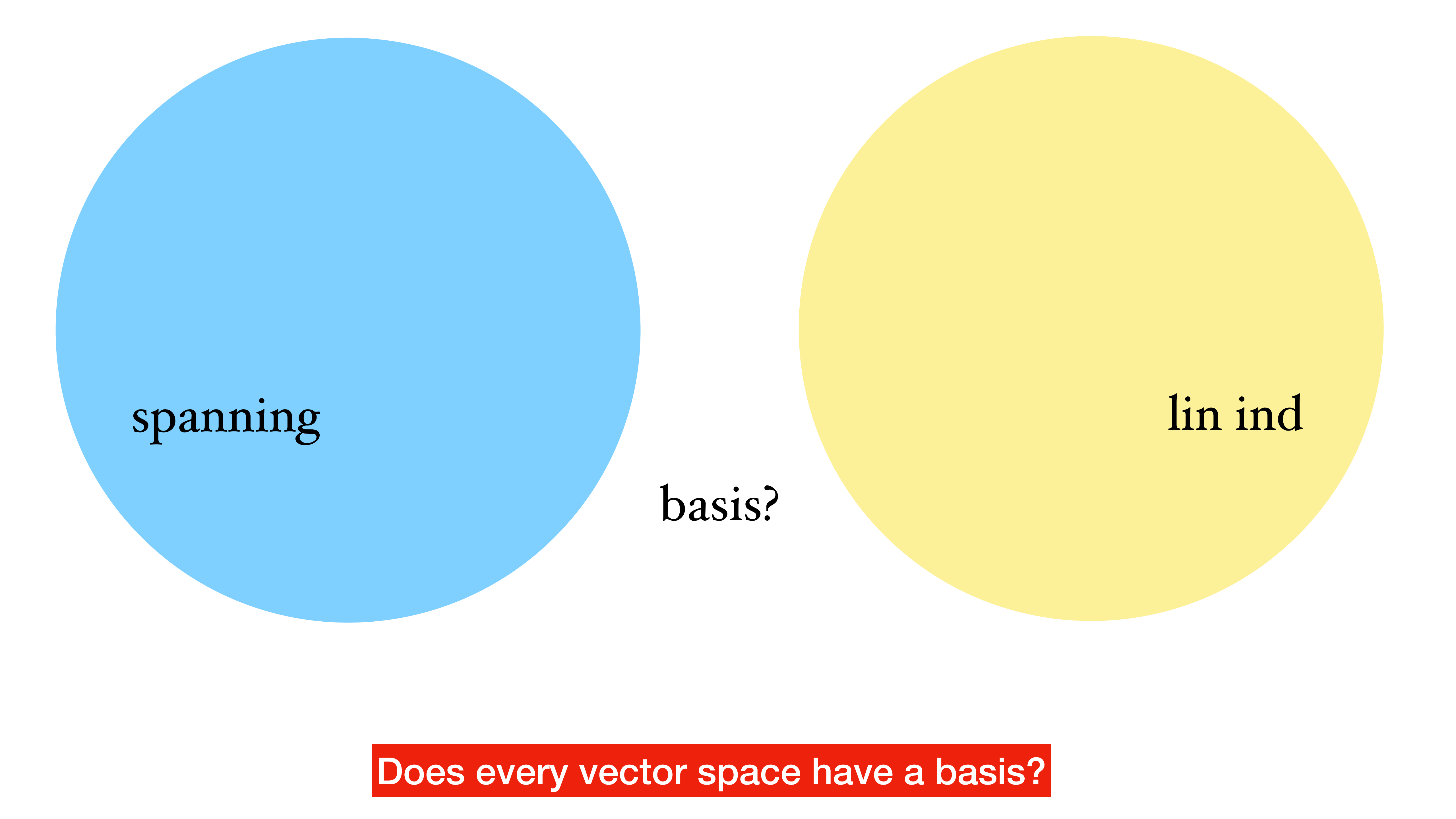
Try those warm-up problems.



spanning

basis

lin ind



spanning

basis?

lin ind

Does every vector space have a basis?

Eventually want to show:

Every vector space with a finite spanning set has a (finite) basis.

Any two finite bases for a vector space have the same cardinality.
(So dimension is well defined.)

Every finite-dimensional vector space over a field $\mathbb{F}$ is isomorphic to some $\mathbb{F}^n$.

(Upshot: we lose no generality by considering the vector spaces $\mathbb{F}^n$.)

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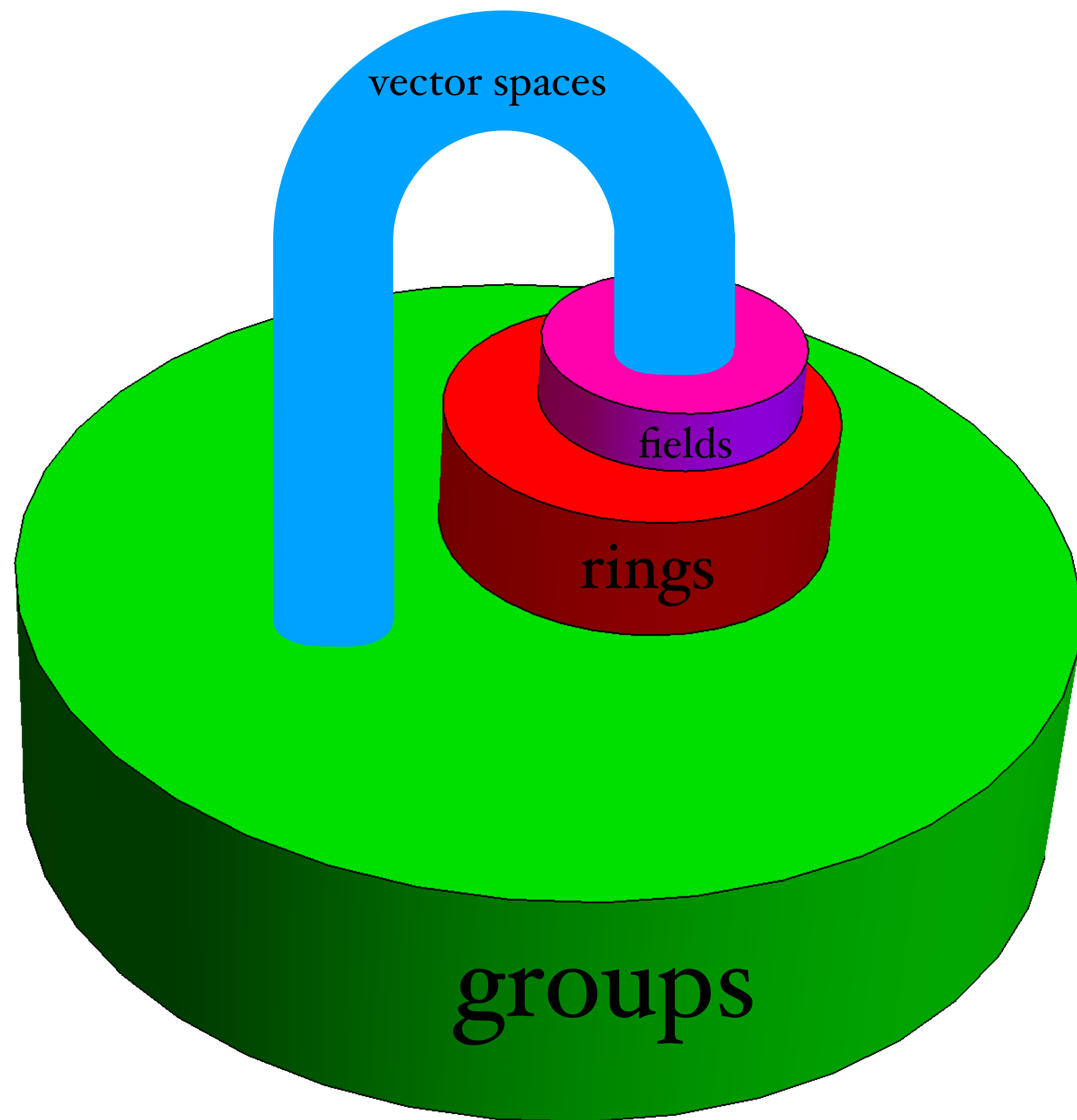
In fact, every vector space has a basis, if we assume the Axiom of Choice; but this does not follow from the definition alone.

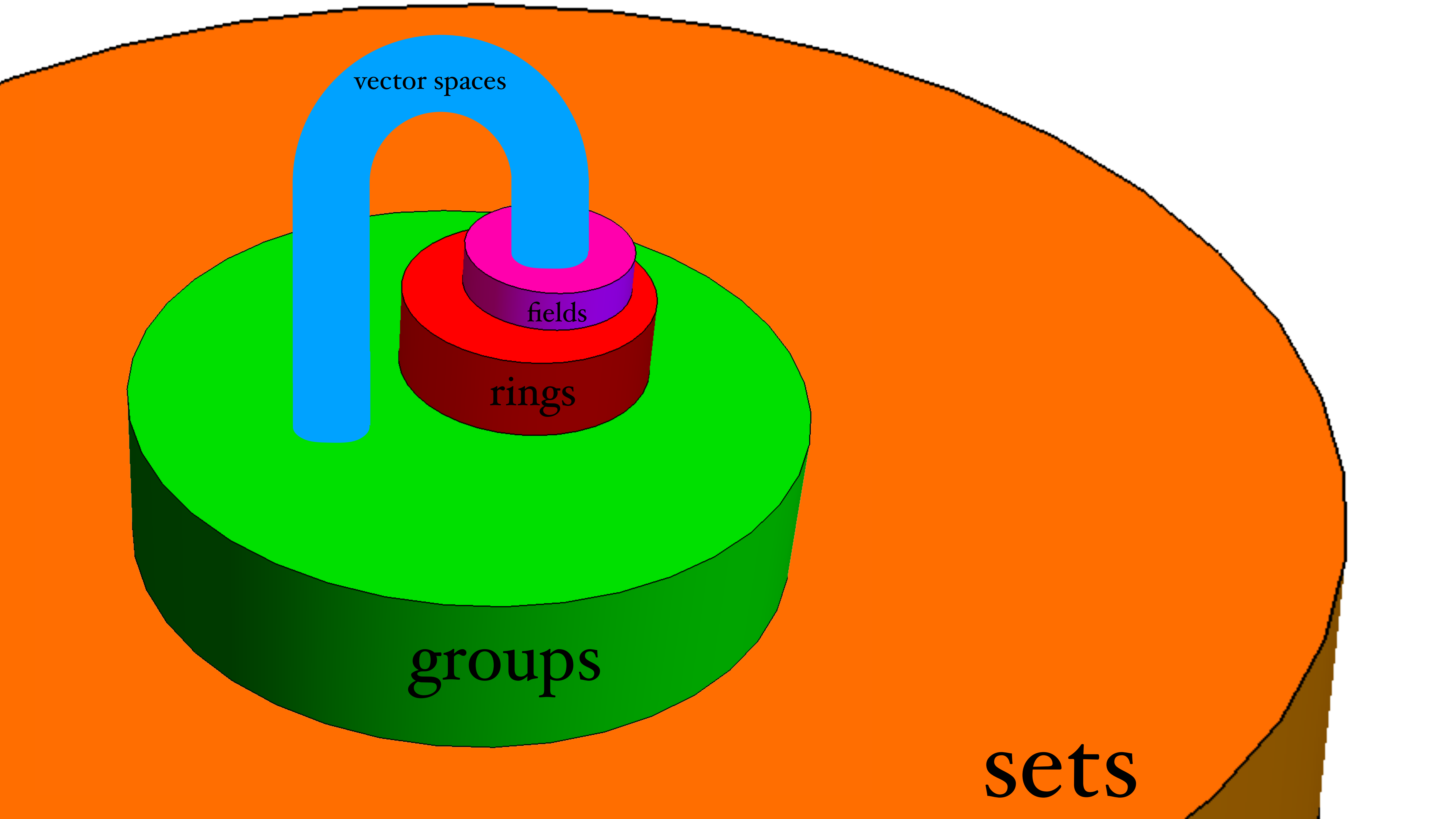
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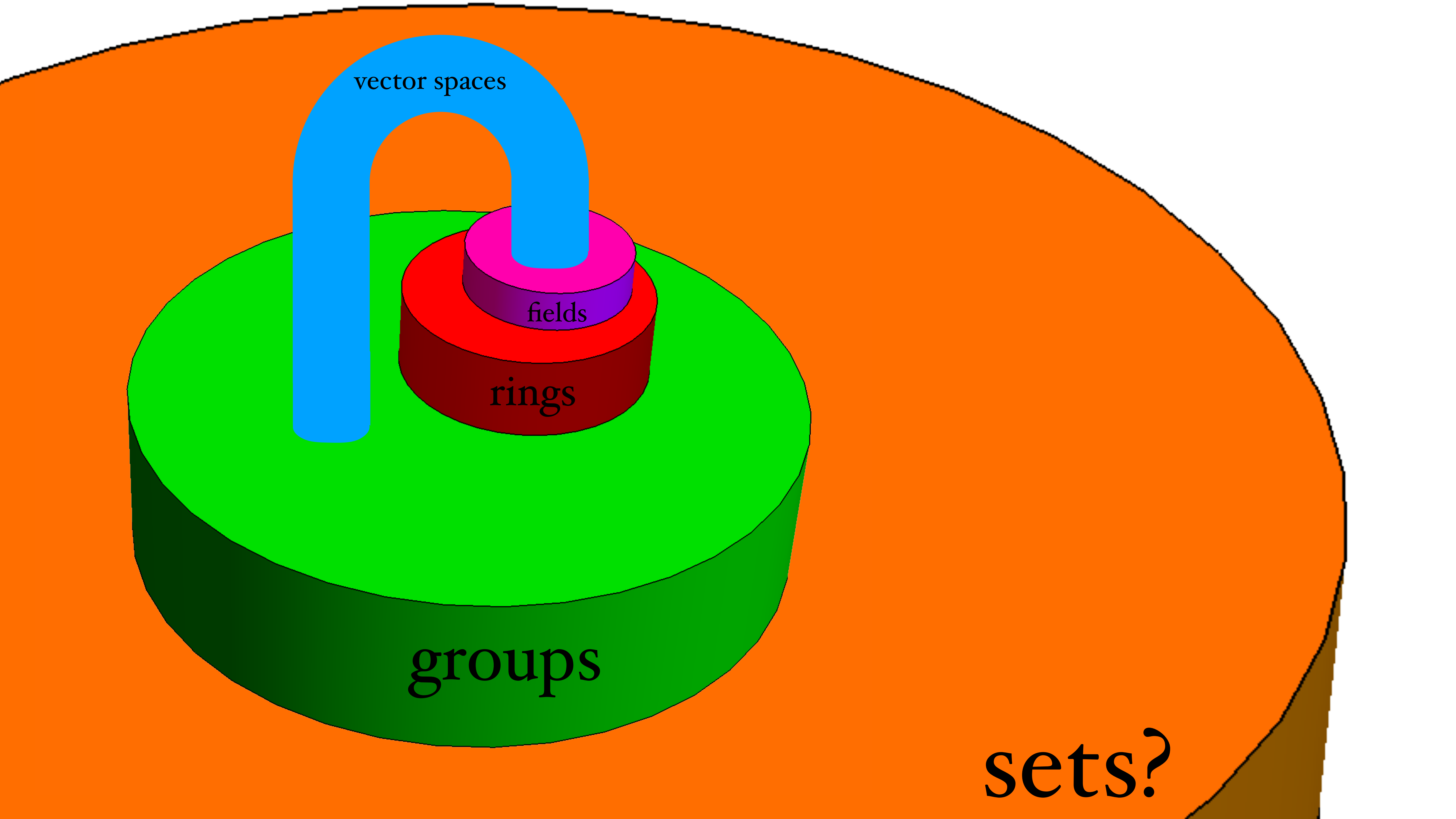
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Today:

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L
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Prop (2.8): Every vector space with a finite spanning set has a (finite) basis.

Proof: Let $v_1, \dots, v_p$ be a finite spanning set for V .

If it's linearly independent, we're done. (It's a basis.)

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But then some v_i with $\alpha_i \neq 0$ is a linear combination of the others:

$$\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_p v_p = -\alpha_i v_i \quad \implies \quad \frac{\alpha_1}{-\alpha_i} v_1 + \frac{\alpha_2}{-\alpha_i} v_2 + \dots + \frac{\alpha_p}{-\alpha_i} v_p = v_i$$

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So v_i can be removed from the set and it still spans.

Now iterate until the result is a basis. (Worst case: remove all vectors, V is the zero v.s.) ■

Composing linear transformations

$$\begin{pmatrix} e & g \\ f & h \end{pmatrix} \begin{pmatrix} a & c \\ b & d \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} =$$

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How many multiplications of field elements
does it take to multiply two 2x2 matrices?
What about 3x3, or nxn?

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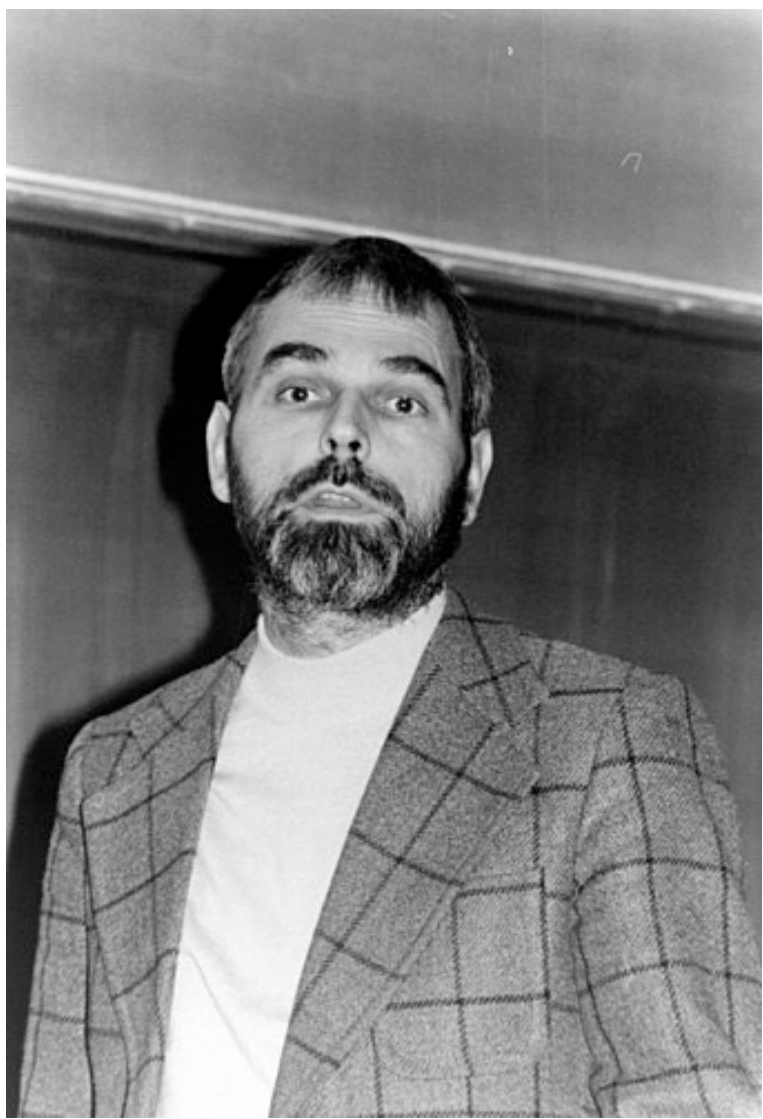
How what is the most efficient way to multiply two $n \times n$ matrices?

The standard algorithm requires n^3 multiplications of field elements.

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Note: multiplication is much more computationally expensive than addition.



Volker Strassen, 1979

Strassen's faster matrix multiplication strategy (1969): $\sim n^{2.81}$ multiplications of scalars

$$\begin{aligned} p1 &= a(f - h) \\ p3 &= (c + d)e \\ p5 &= (a + d)(e + h) \\ p7 &= (a - c)(e + f) \end{aligned}$$

$$\begin{aligned} p2 &= (a + b)h \\ p4 &= d(g - e) \\ p6 &= (b - d)(g + h) \end{aligned}$$

The A x B can be calculated using above seven multiplications.
Following are values of four sub-matrices of result C

a	b
c	d

A

x

e	f
g	h

B

=

p5 + p4 - p2 + p6	p1 + p2
p3 + p4	p1 + p5 - p3 - p7

C

Current fast matrix multiplication record:

(October 2020): $\sim n^{2.372}$ multiplications of field elements



Virginia Vassilevska Williams

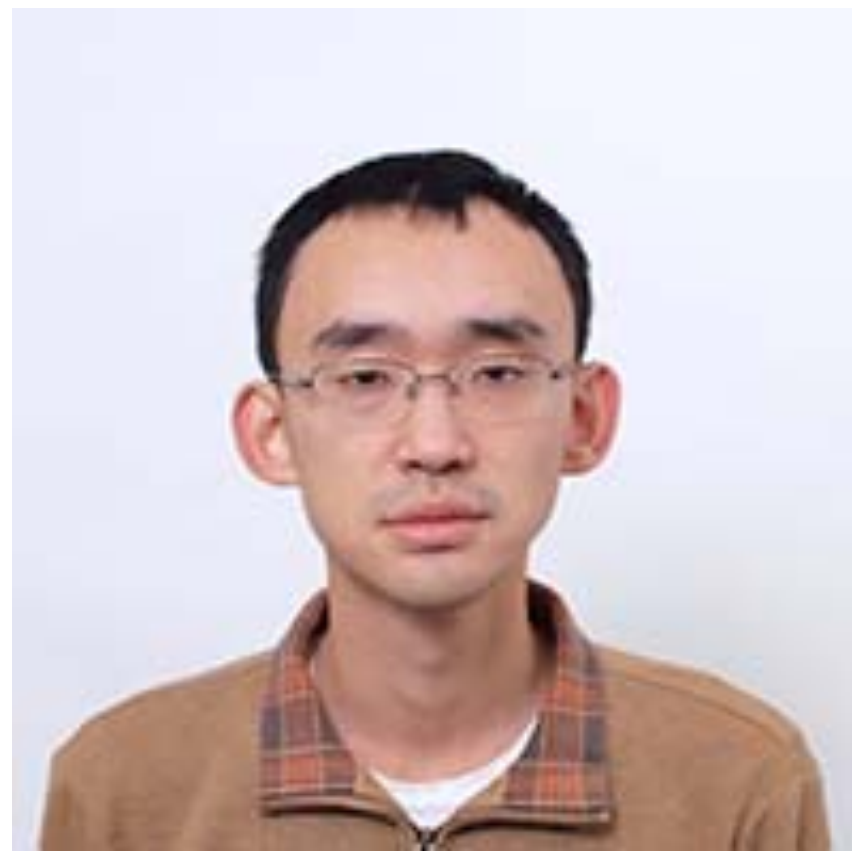


Josh Alman

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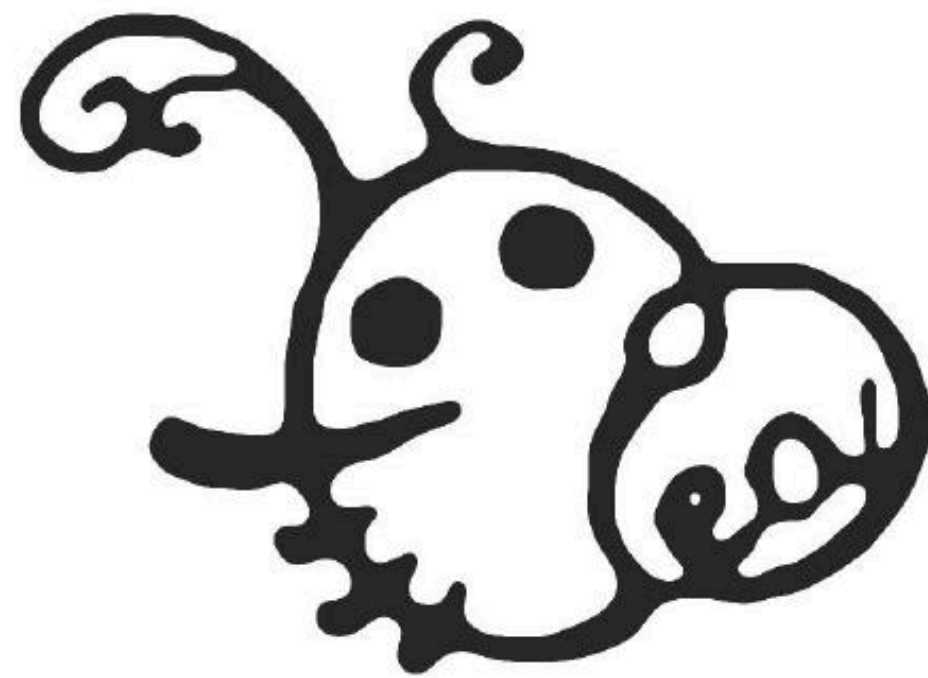
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Ran Duan



Hongxun Wu



Renfei Zhou



Virginia Vassilevska Williams



Josh Alman

I.5

In general, matrix multiplication is not commutative:

except under special conditions, $AB \neq BA$

I.5

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In fact, given matrices A and B , AB may be defined, but BA not!

$$A : \mathbb{R}^3 \rightarrow \mathbb{R}^4$$

$$B : \mathbb{R}^2 \rightarrow \mathbb{R}^3$$

$$AB : \mathbb{R}^2 \rightarrow \mathbb{R}^4$$

$BA :$ not a well-defined
function

I.5

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Both the determinant and the trace of a matrix are field elements.

They are also invariants of the underlying linear transformation: they don't depend on a choice of basis!

1.5

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Matrices may not commute, but their “shadows” always do.

1.5

Why do we have $\text{tr}(AB) = \text{tr}(BA)$?

(hint. the 2x2 case)

$$\begin{pmatrix} e & g \\ f & h \end{pmatrix} \begin{pmatrix} a & c \\ b & d \end{pmatrix} = \begin{pmatrix} ae + bg & ec + gd \\ af + bh & fc + hd \end{pmatrix}$$

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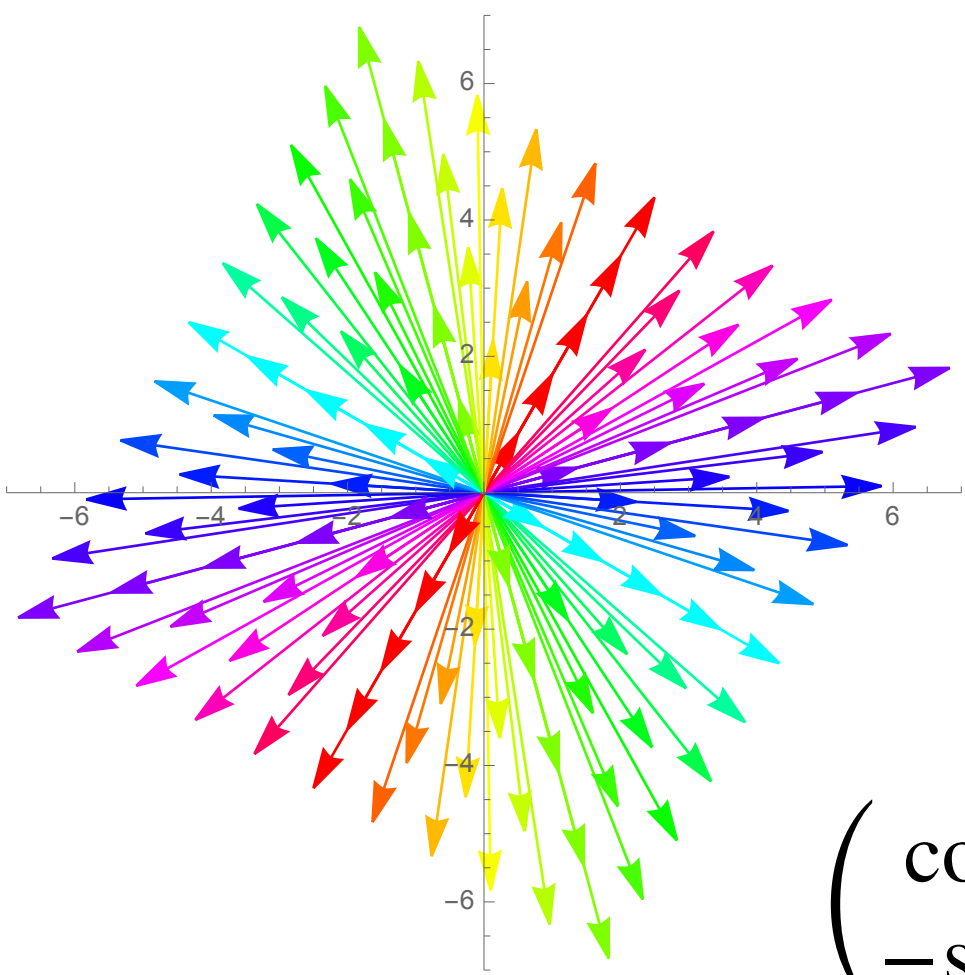
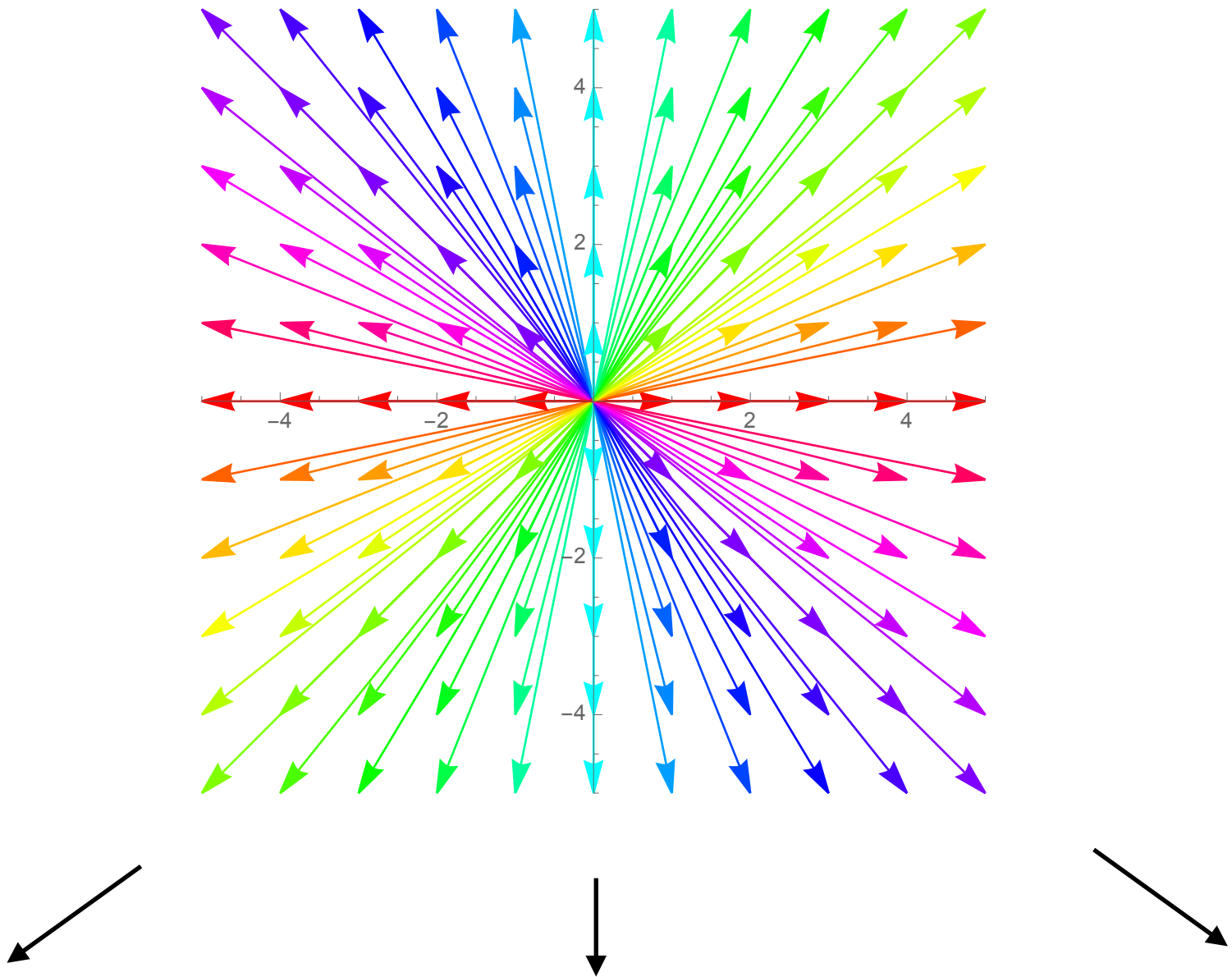
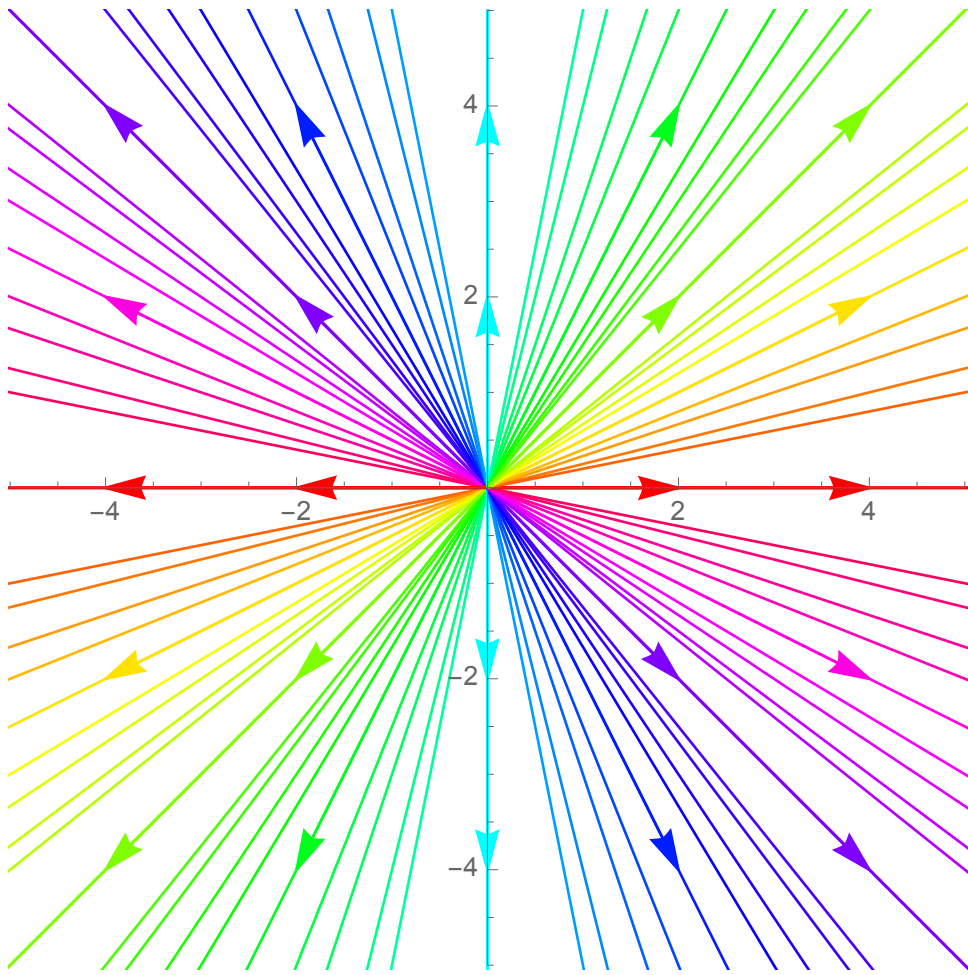
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The two sums along the diagonal are equal: just a rearrangement of terms.

Invertible linear transformations

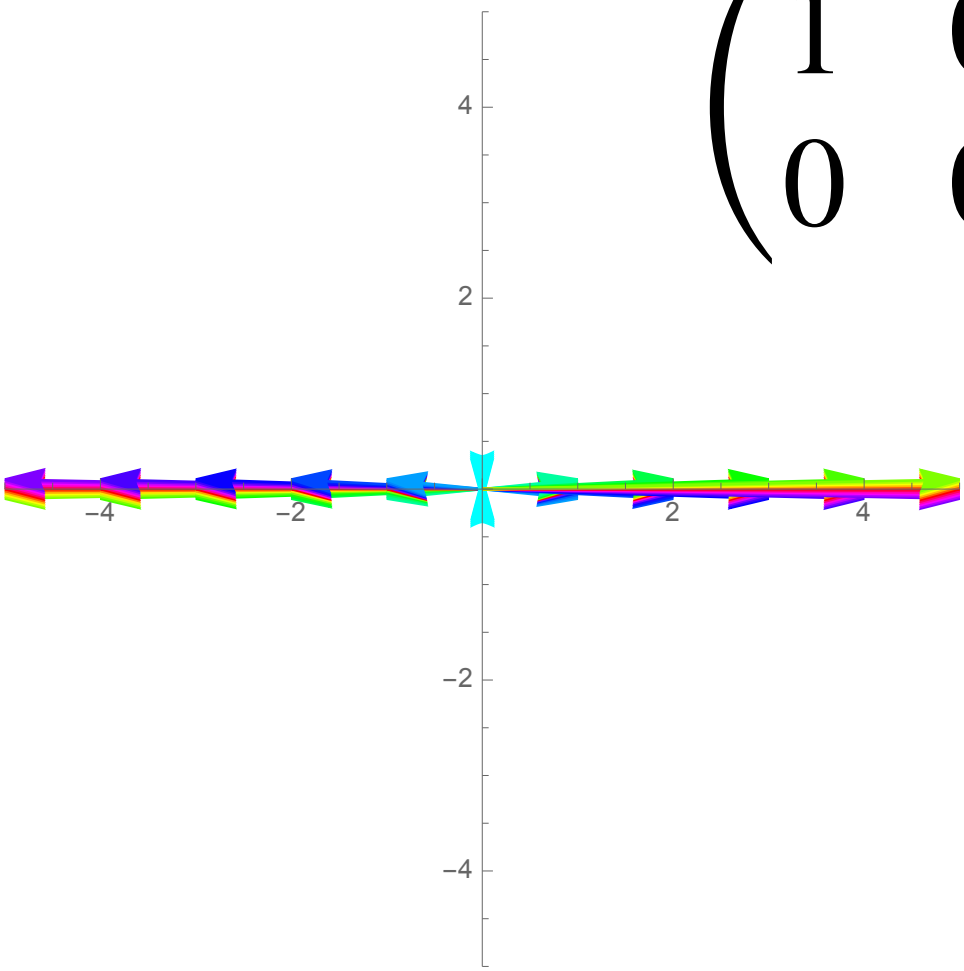
$$\mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$$



$$\begin{pmatrix} \cos \pi/3 & \sin \pi/3 \\ -\sin \pi/3 & \cos \pi/3 \end{pmatrix}$$

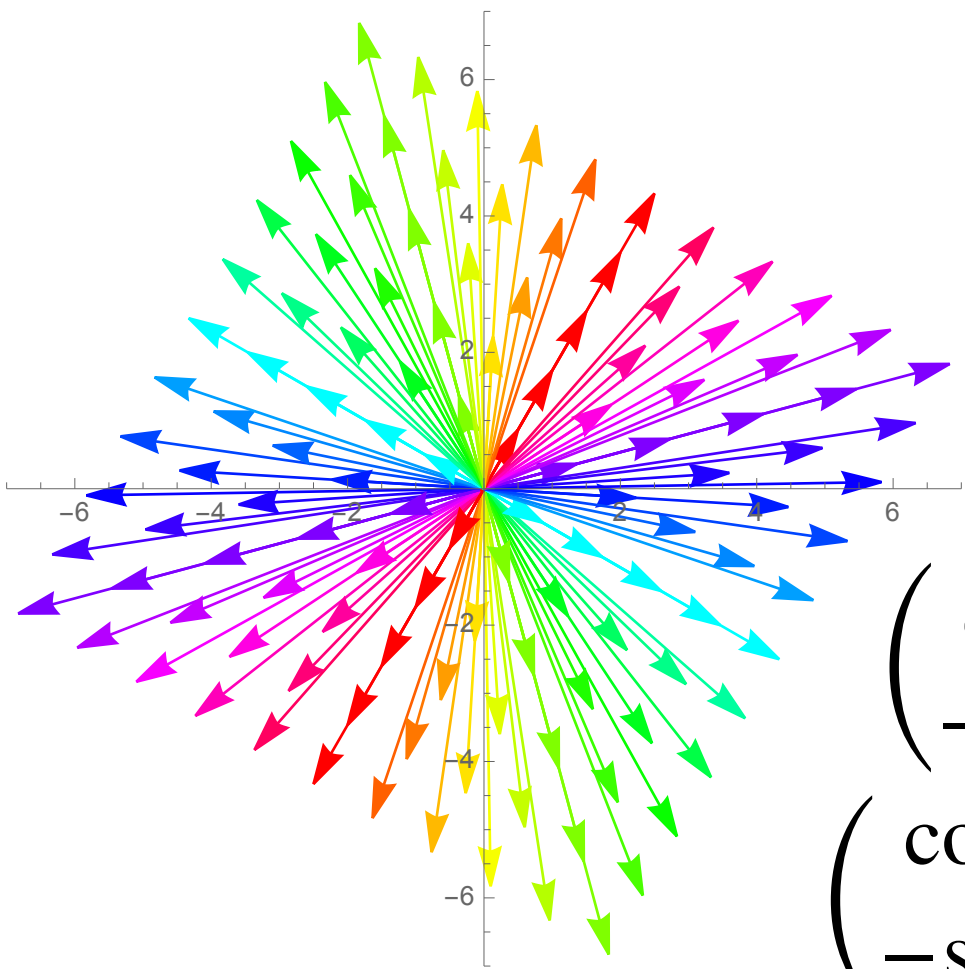
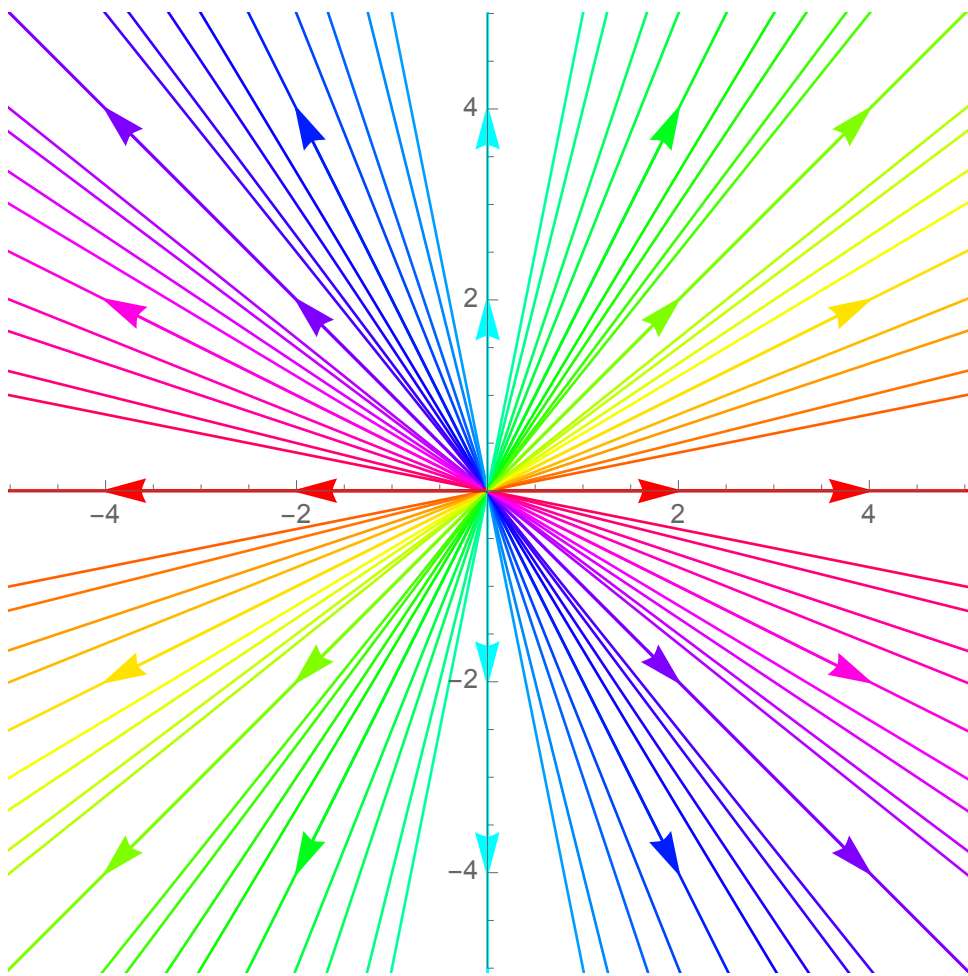
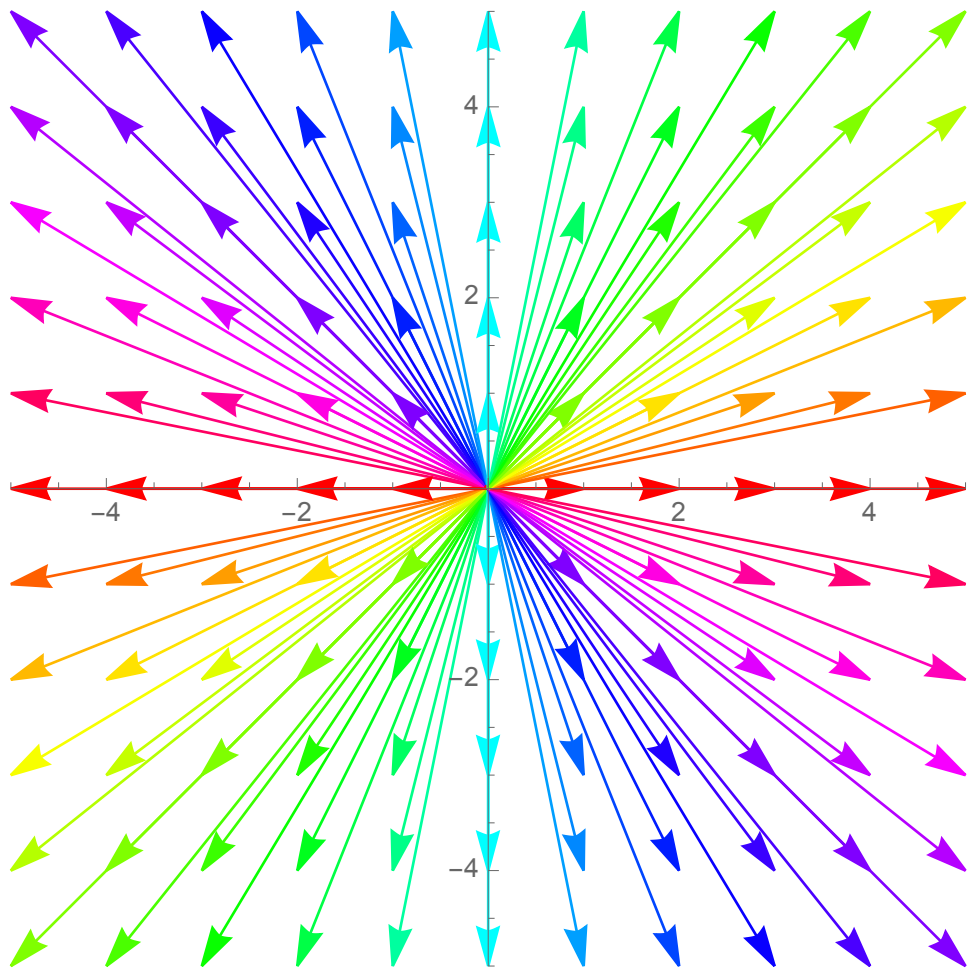
$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$



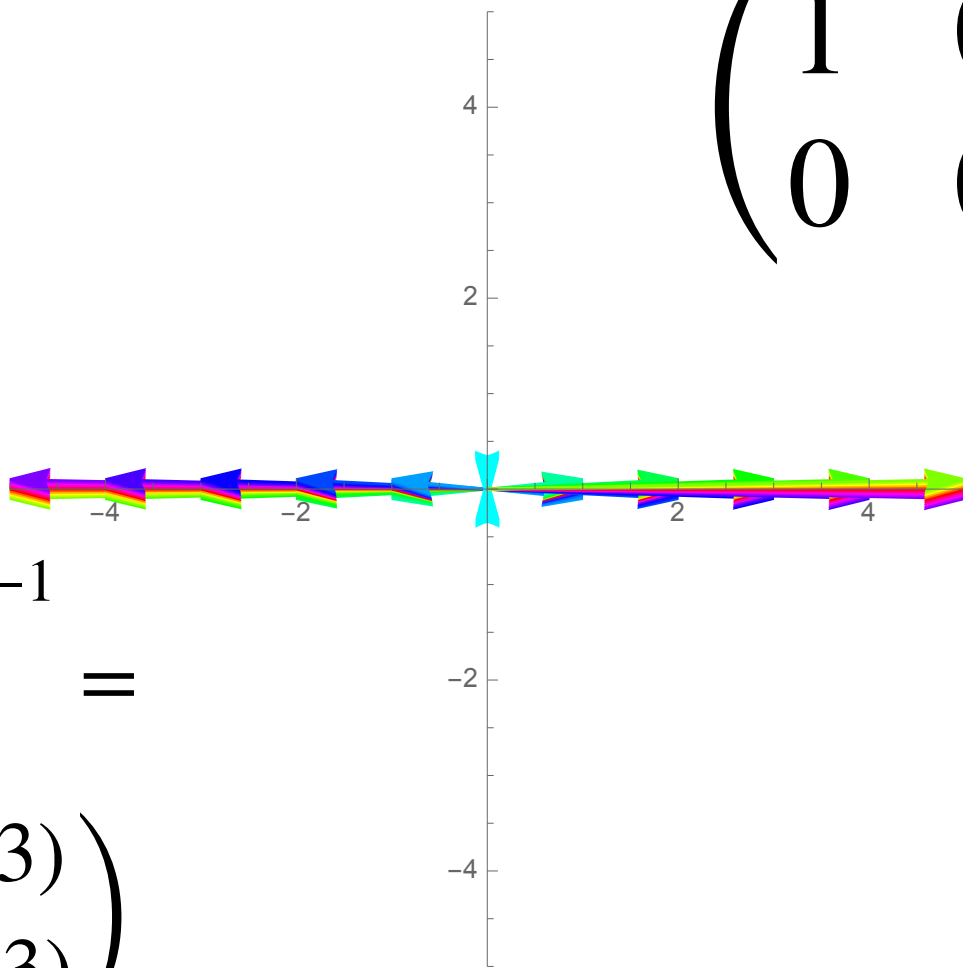
Invertible linear transformations

$$\mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}^{-1} = \begin{pmatrix} 1/2 & 0 \\ 0 & 1/2 \end{pmatrix}$$



$$\begin{pmatrix} \cos \pi/3 & \sin \pi/3 \\ -\sin \pi/3 & \cos \pi/3 \end{pmatrix}^{-1} = \begin{pmatrix} \cos(-\pi/3) & \sin(-\pi/3) \\ -\sin(-\pi/3) & \cos(-\pi/3) \end{pmatrix}$$

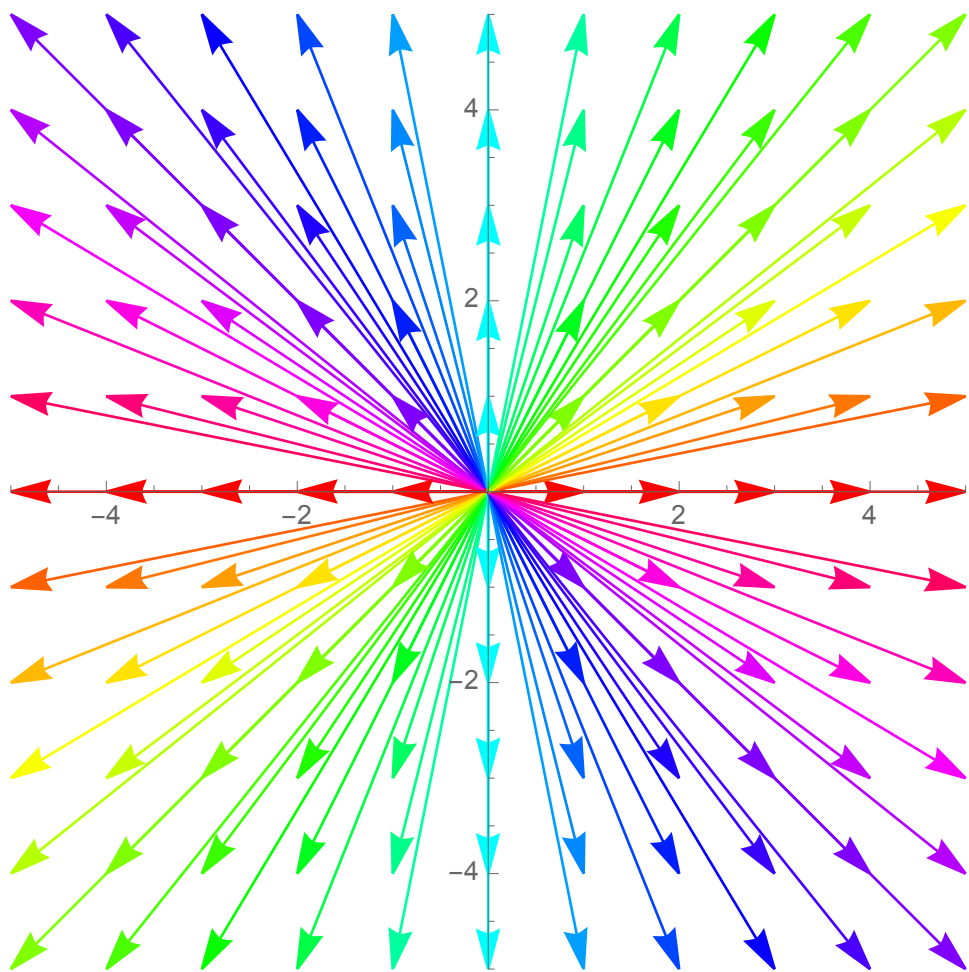


$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

Invertible linear transformations

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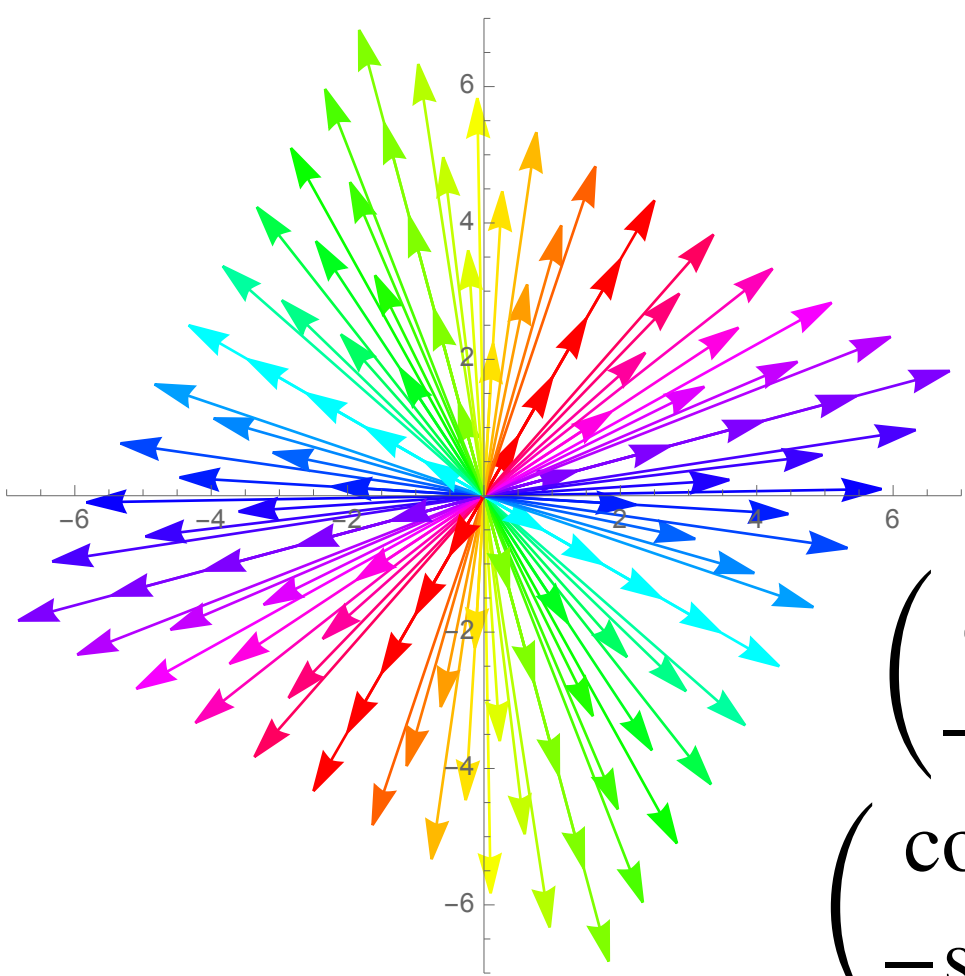
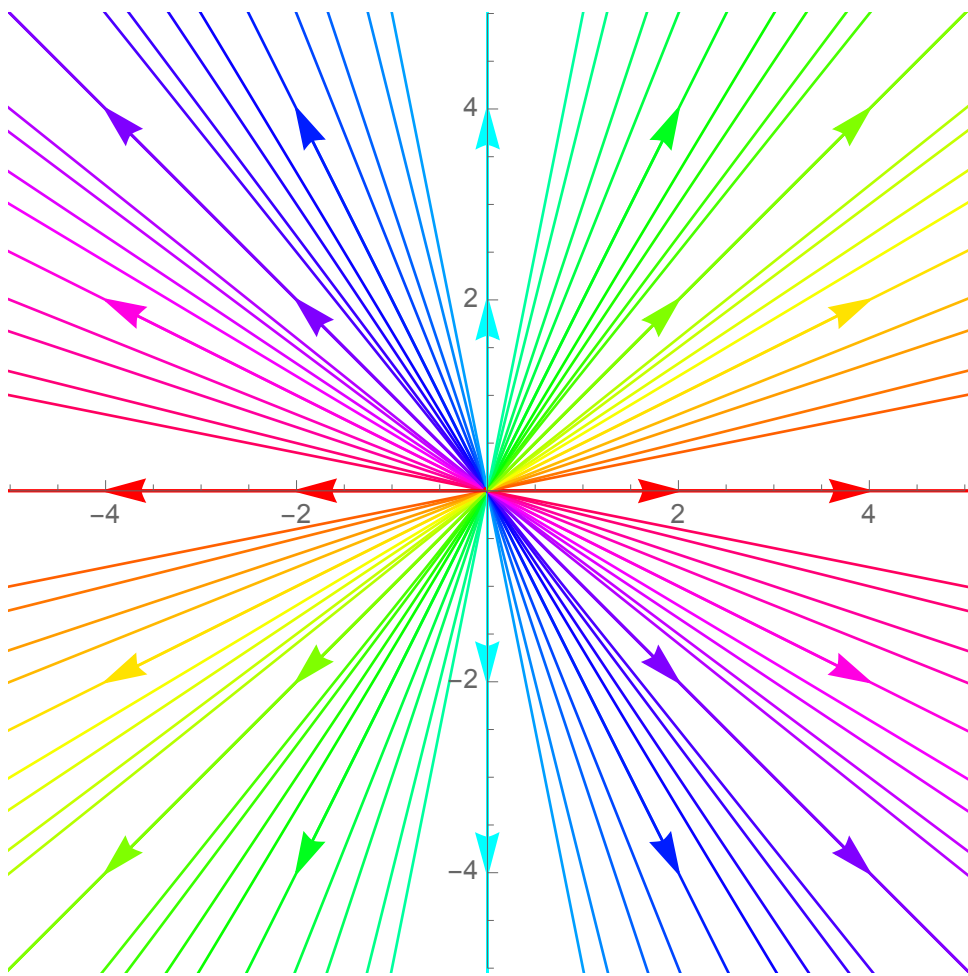
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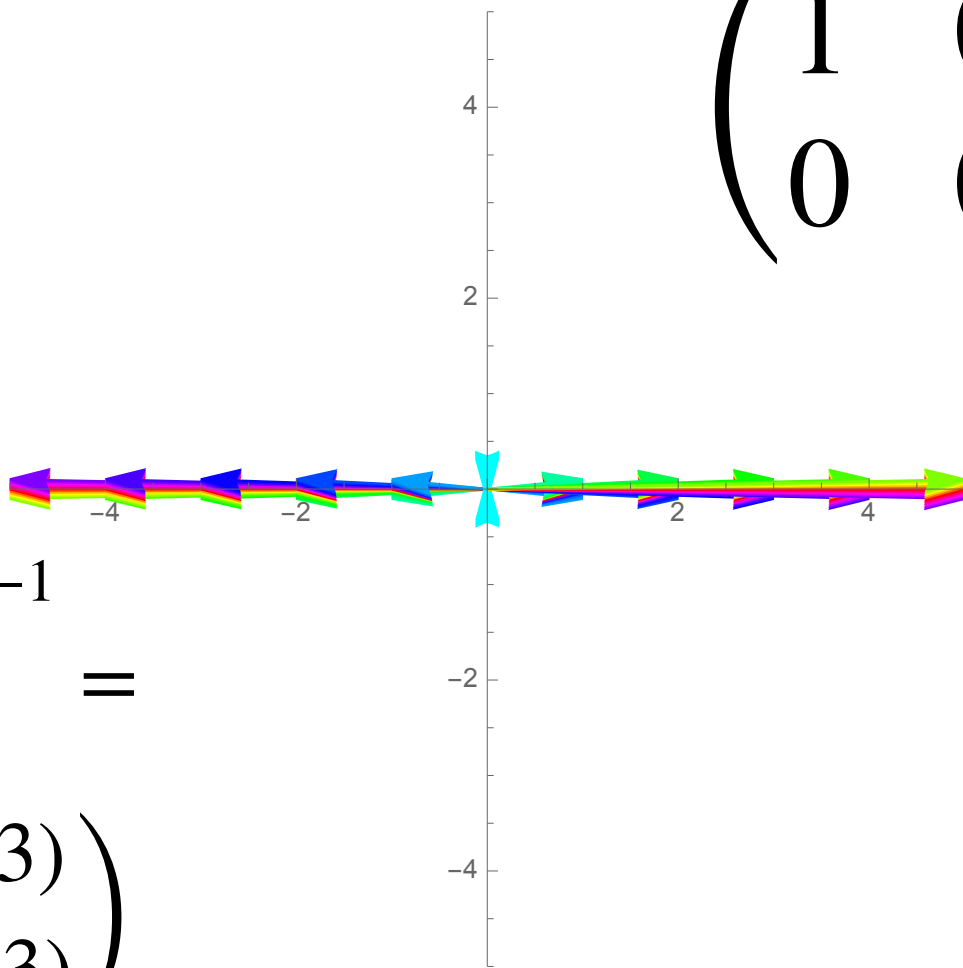
not injective
not surjective



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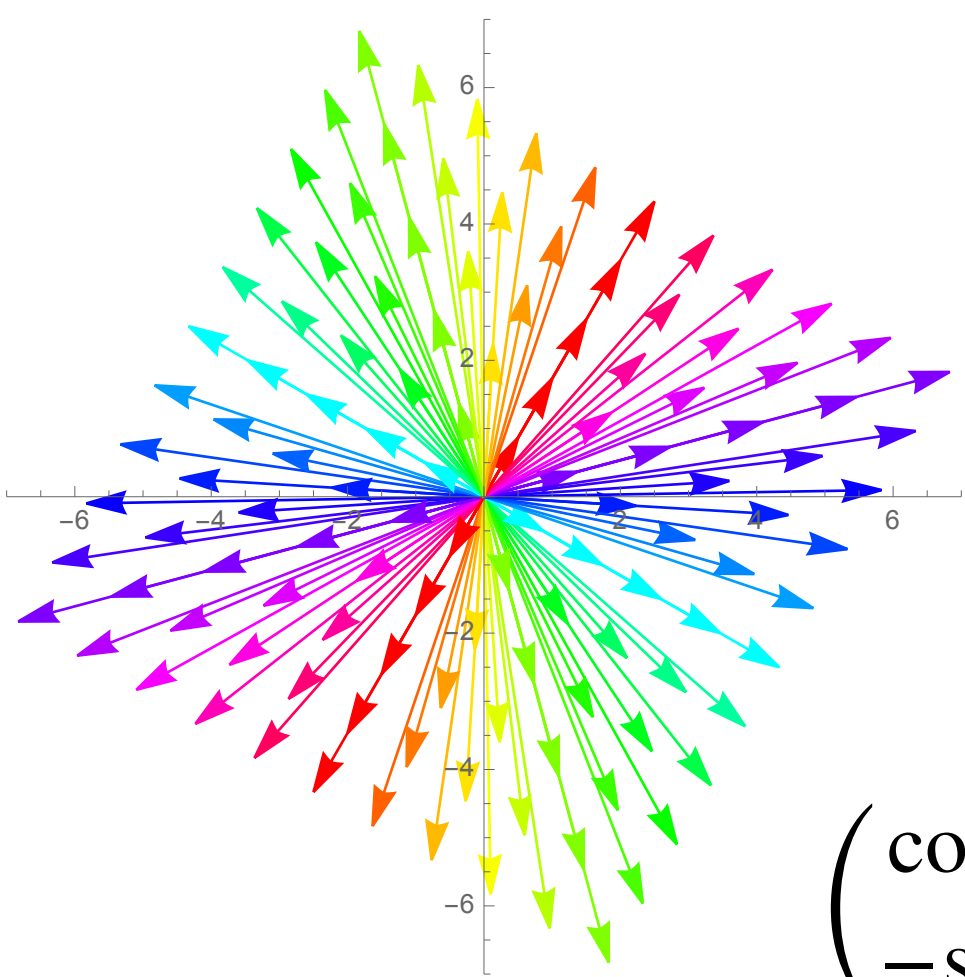
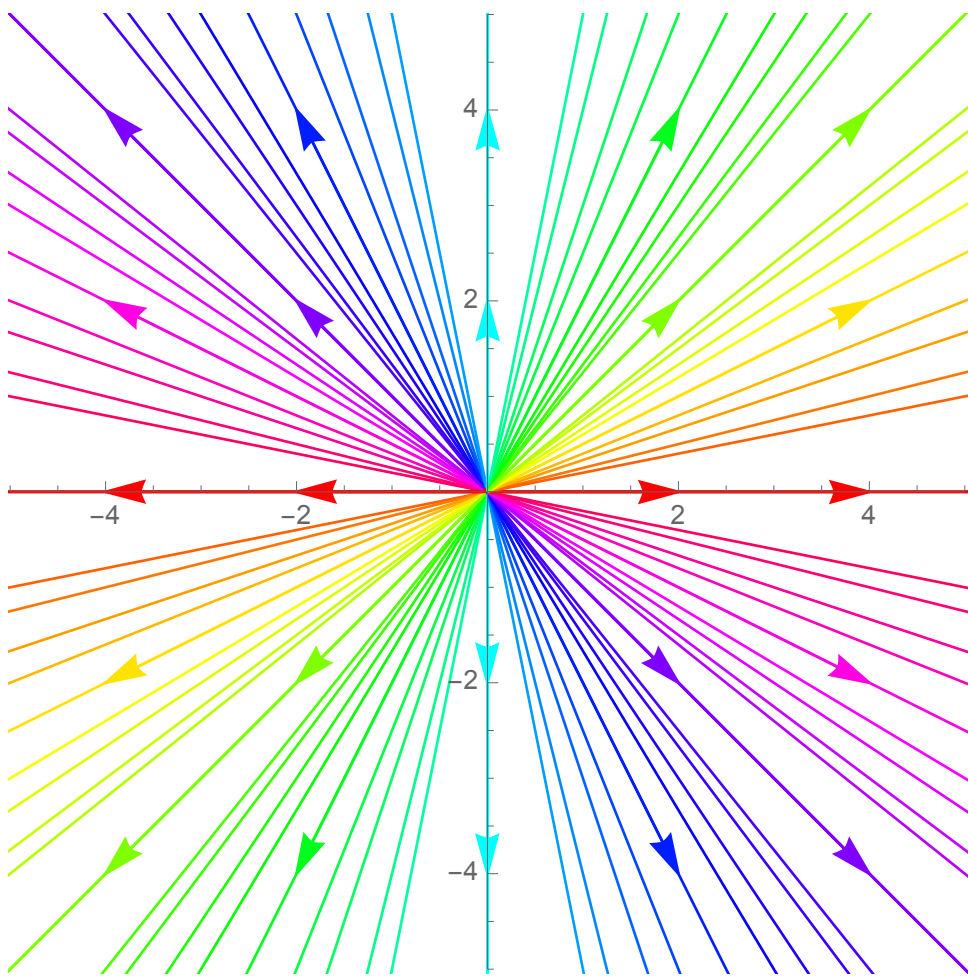
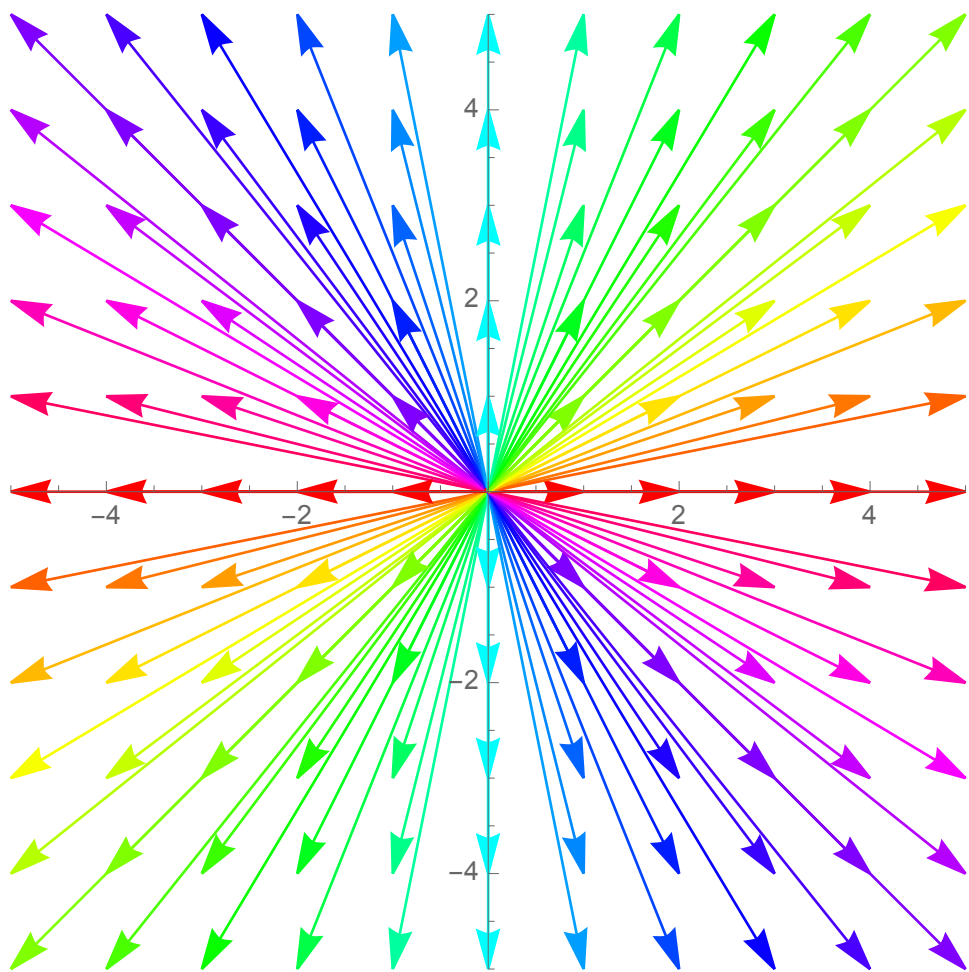
Invertible linear transformations

$$\mathbb{R}^2 \rightarrow \mathbb{R}^2$$

How can you tell when
a linear transformation
or matrix is invertible?

Invertible Matrix Theorem

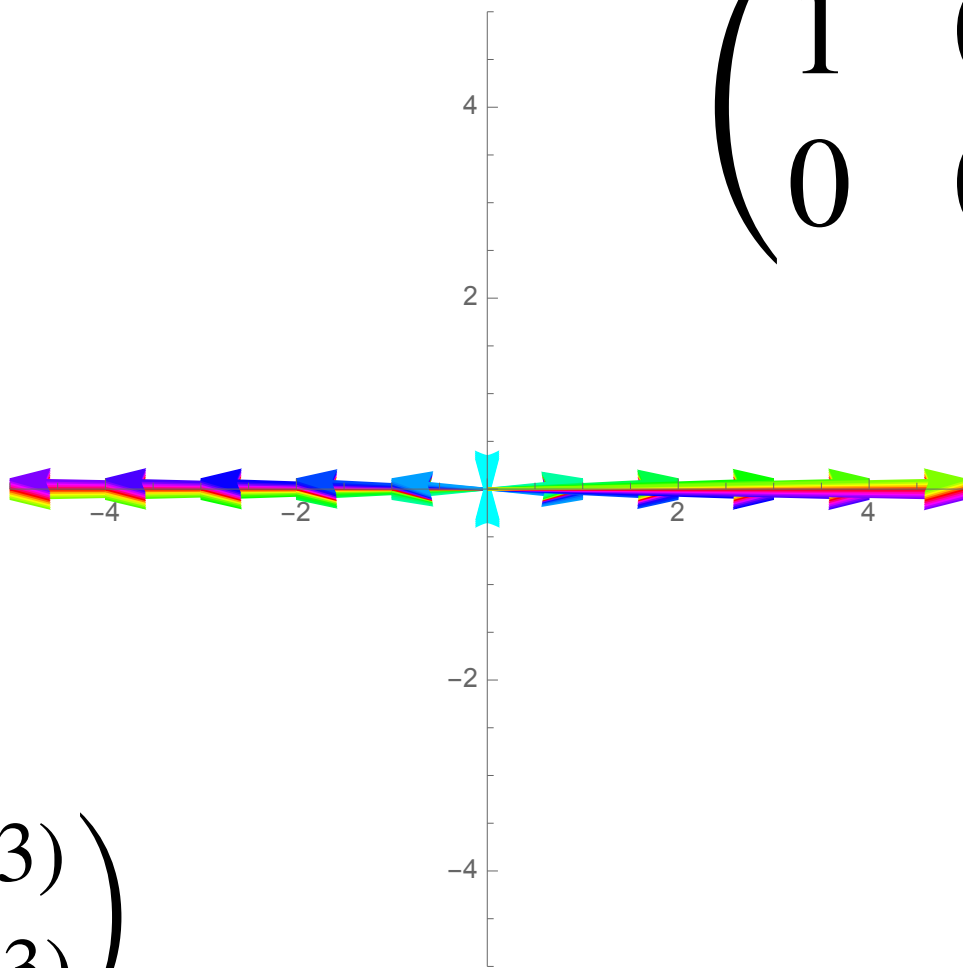
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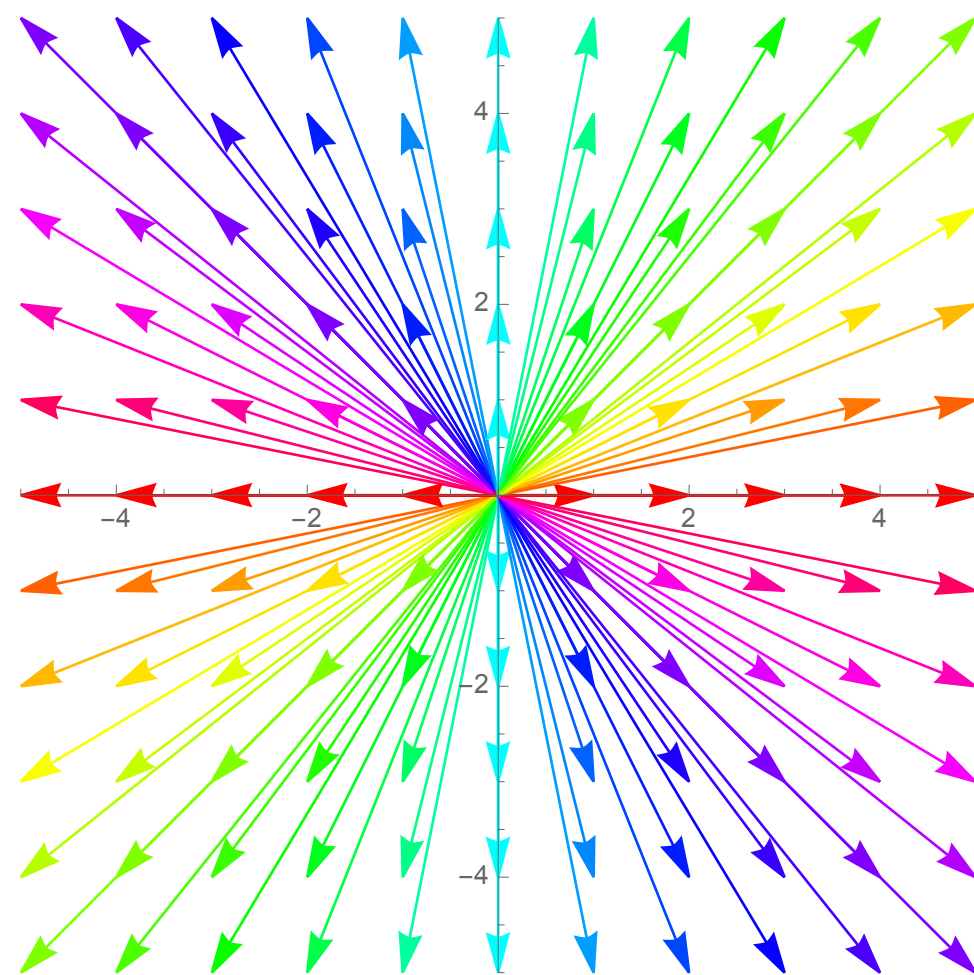
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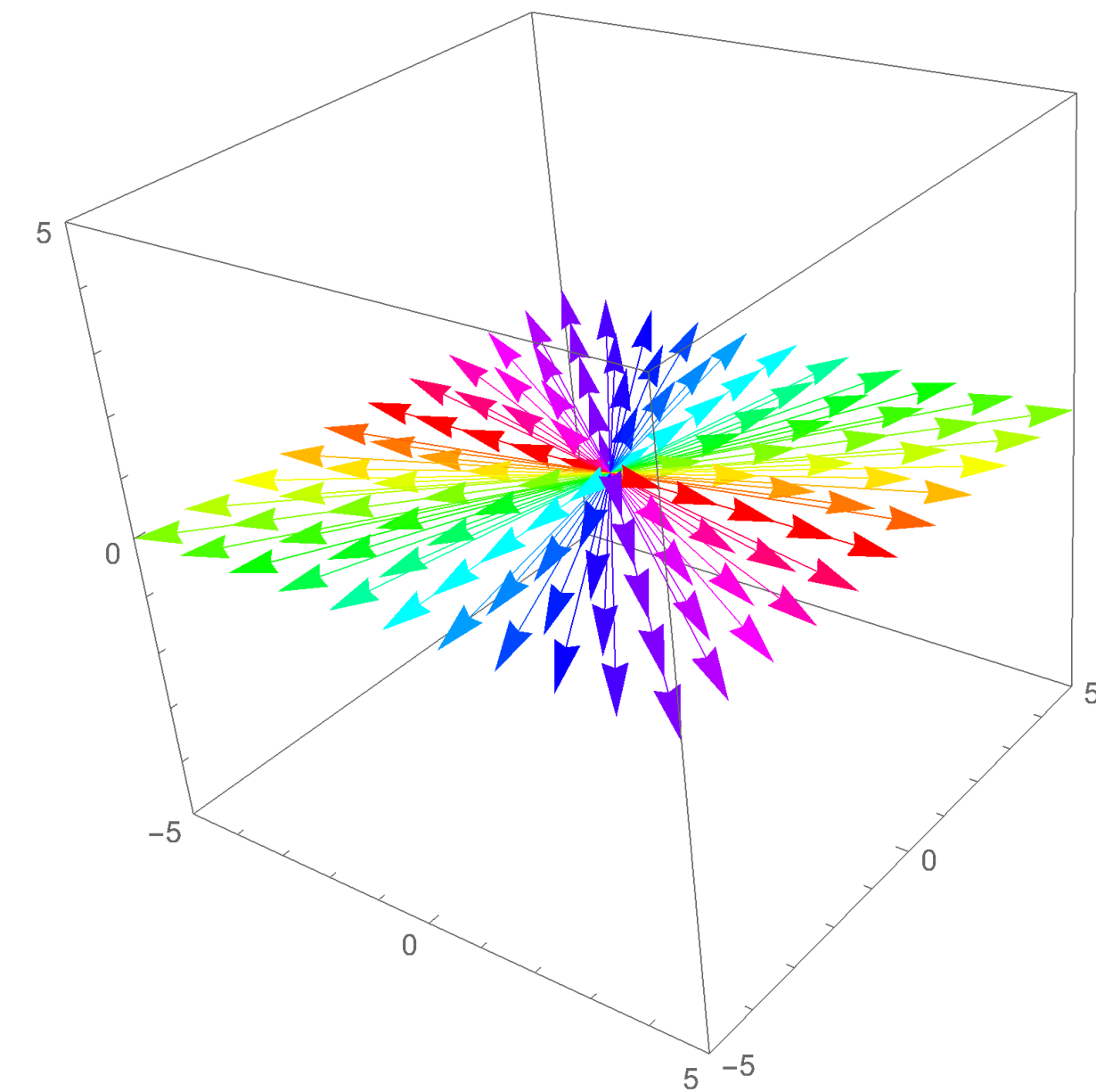
I.6

Invertible linear transformations

$$\mathbb{R}^2 \rightarrow \mathbb{R}^3$$



f



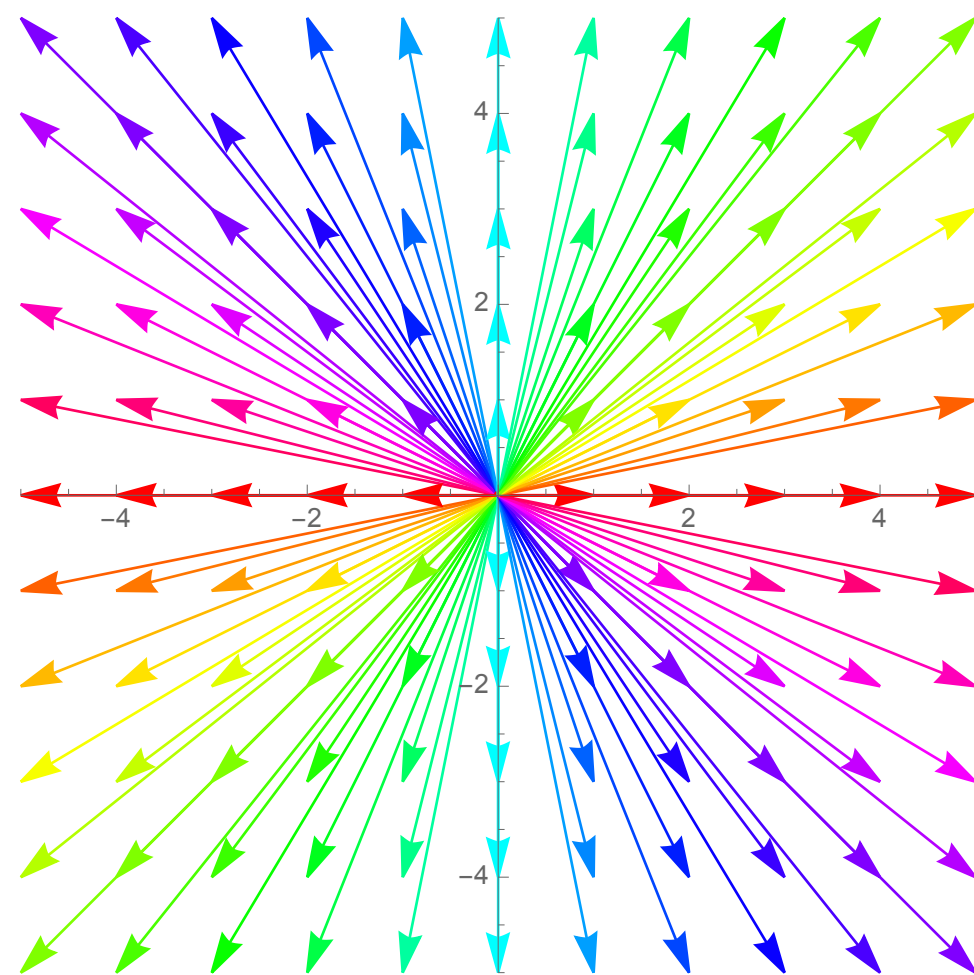
f has a left inverse, but not a right inverse.

a “post” inverse

I.6

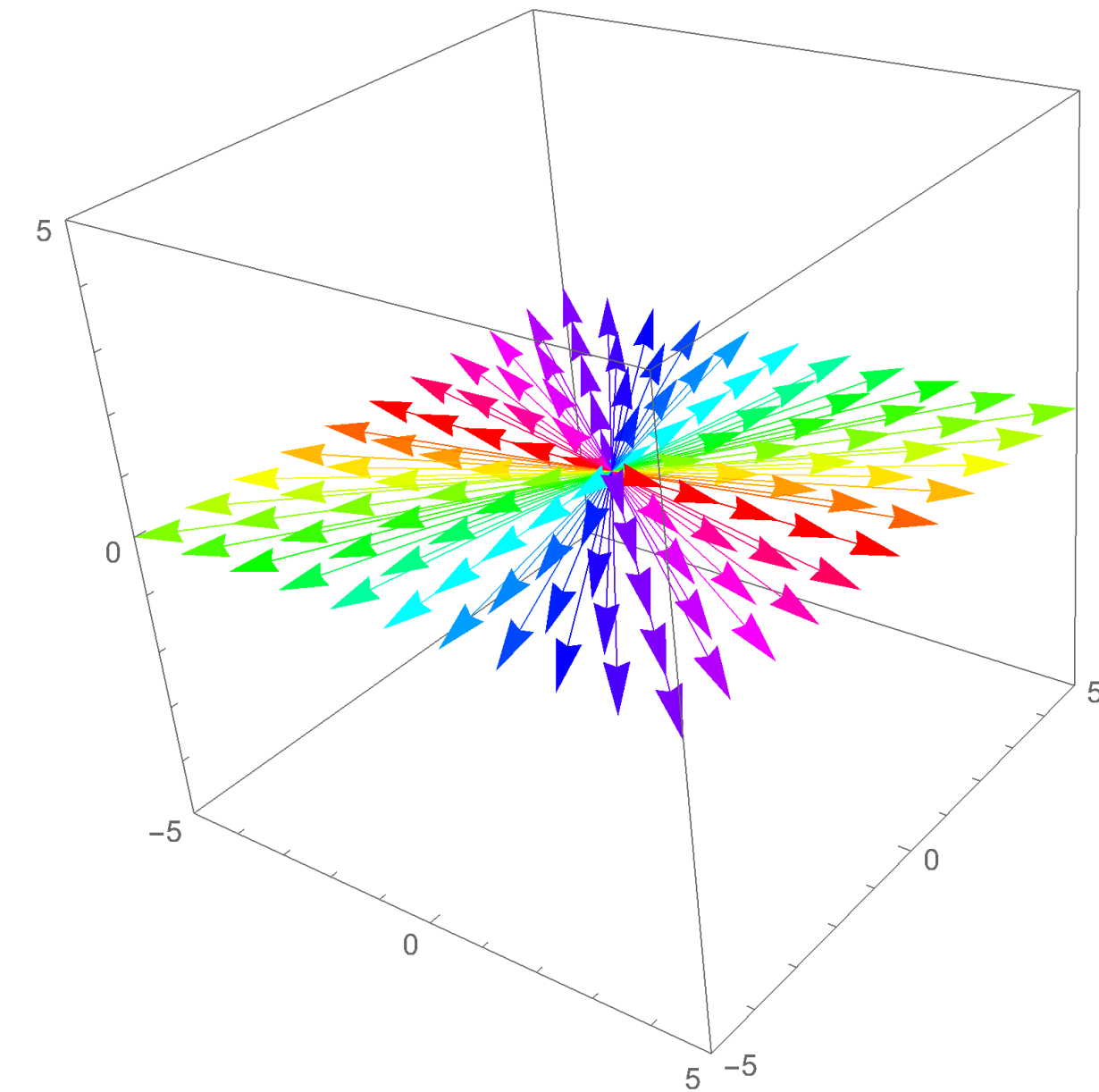
Invertible linear transformations

$$\mathbb{R}^2 \rightarrow \mathbb{R}^3$$



g

←

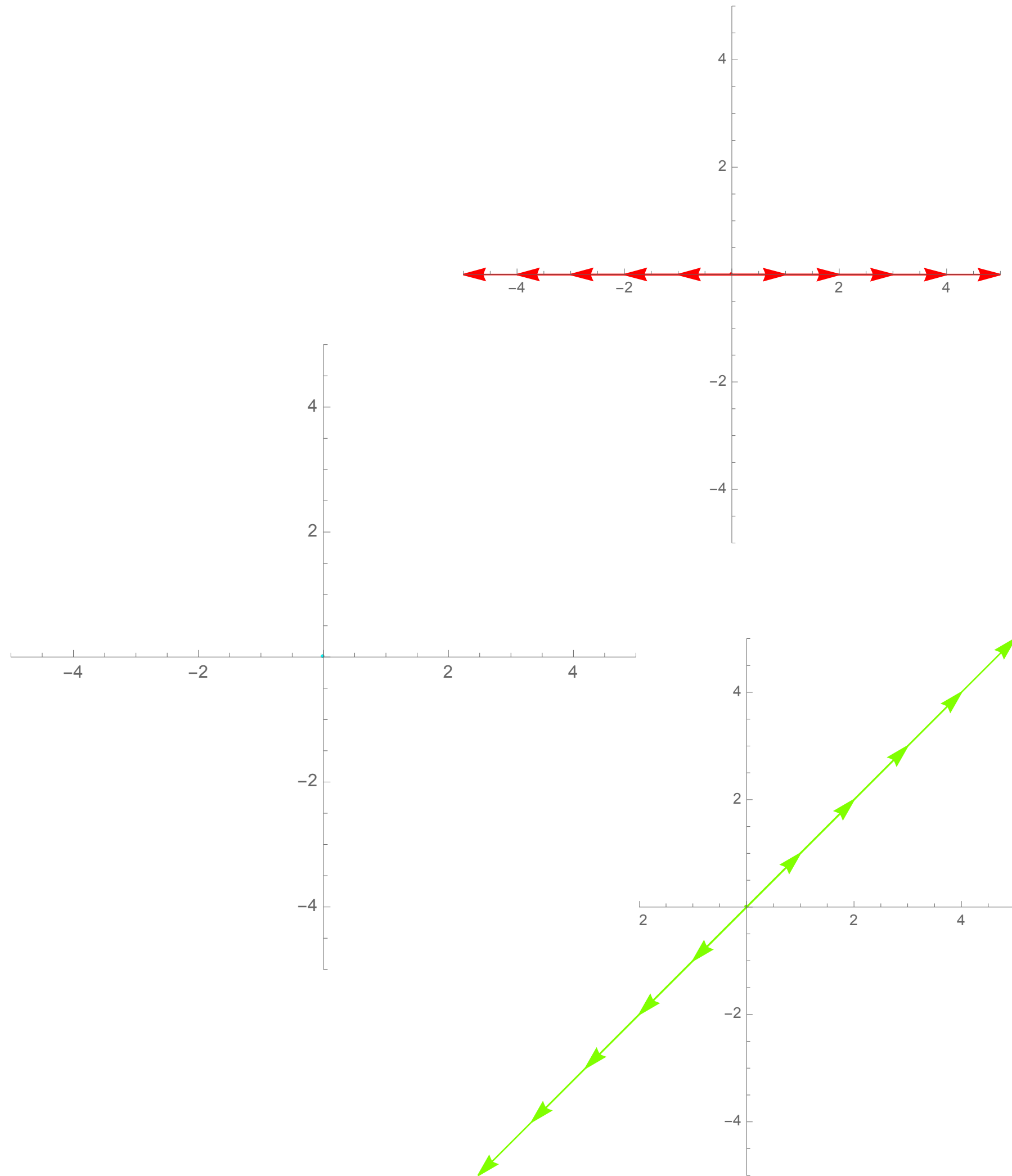


g has a right inverse, but not a left inverse.

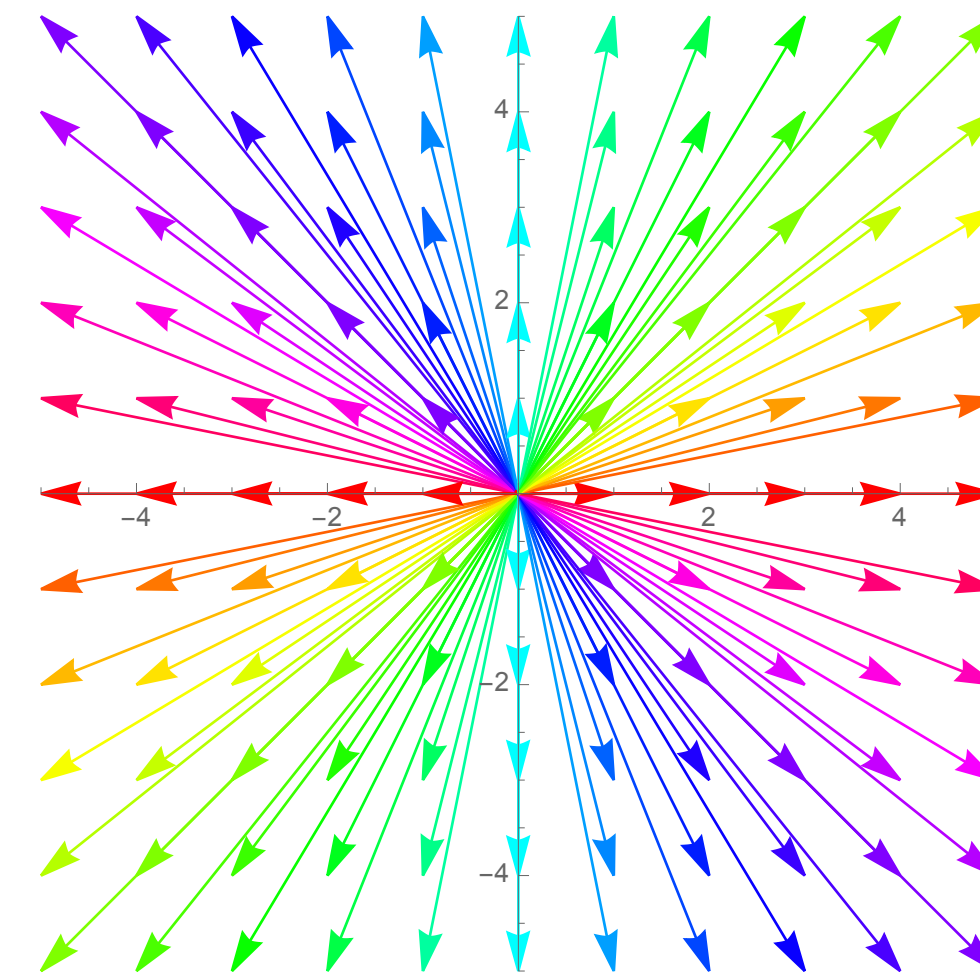
a “pre” inverse

I.7

Subspaces



$\subseteq$



Subspaces need to satisfy the vector space axioms.

But only need to check:

contains zero vector &

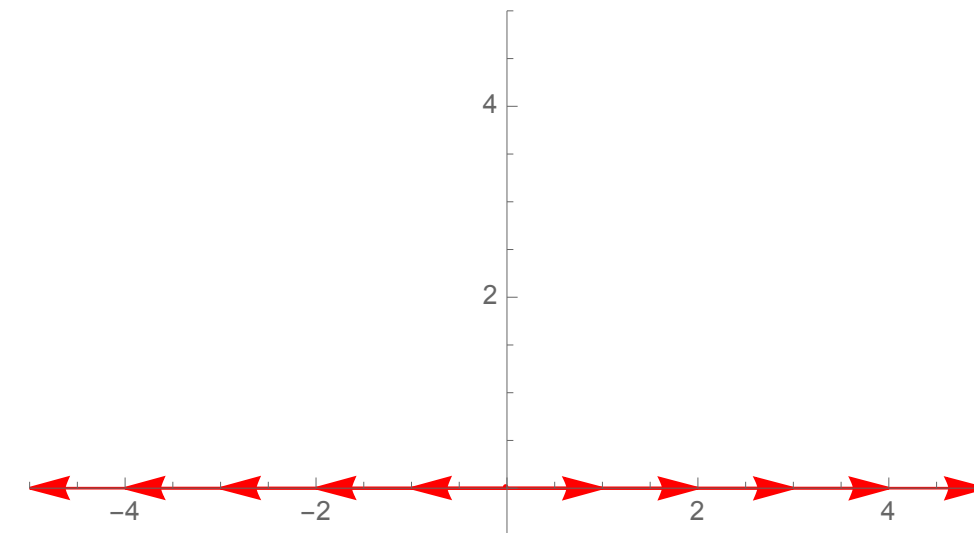
closed under linear combinations

The rest are inherited from the ambient space.

I.7

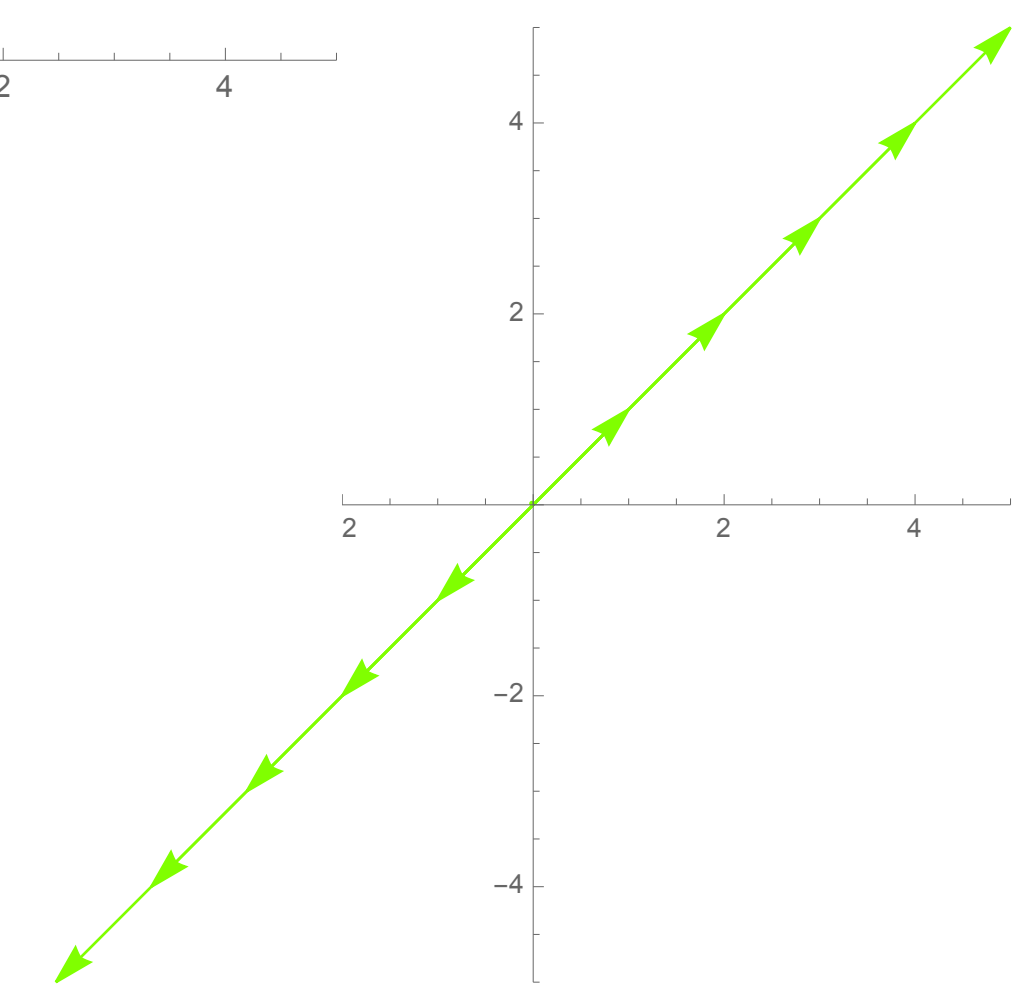
Subspaces

spans

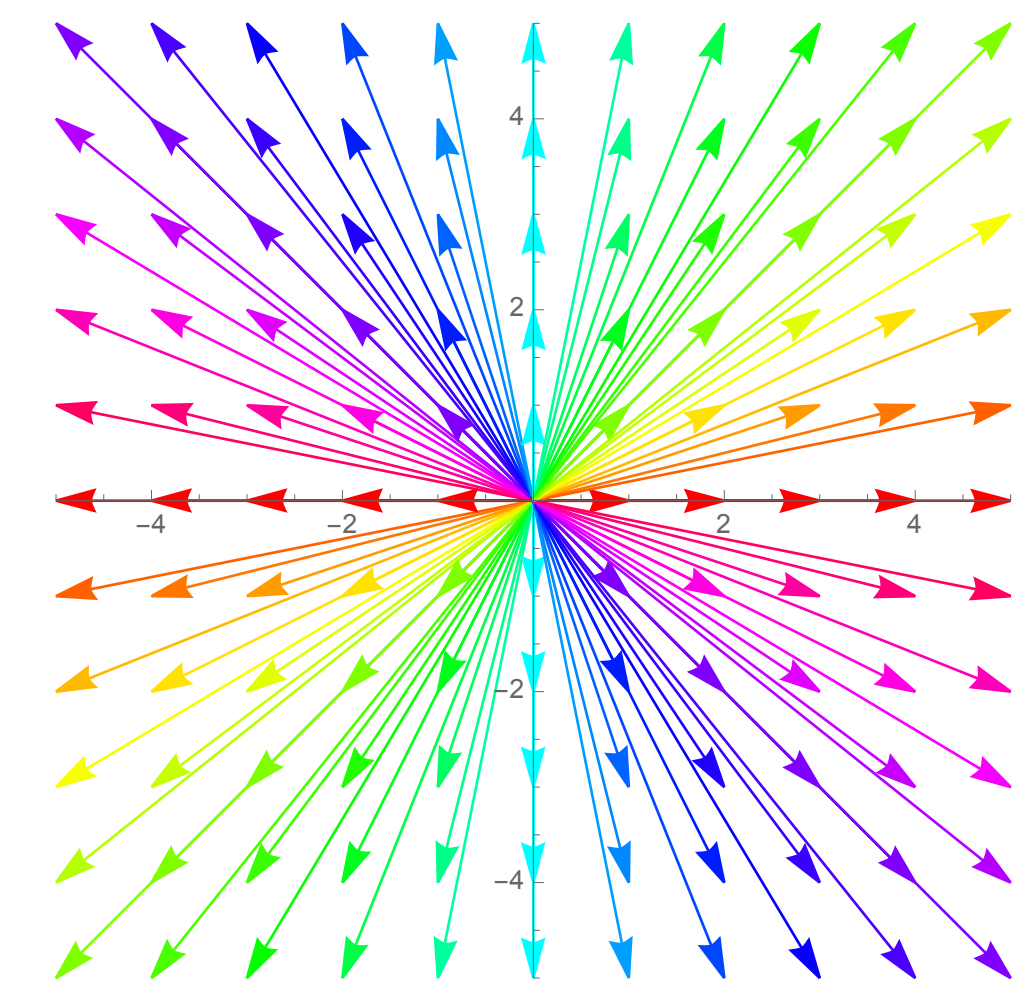


nullspaces

(solution sets
to homogeneous
linear
systems)



$\subseteq$



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contains zero vector &
closed under linear combinations
The rest are inherited from the ambient space.